5G/6G Communication Systems- ECA5601(LAB)

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Experiment 10: 5G Network Slicing Environment Using MATLAB

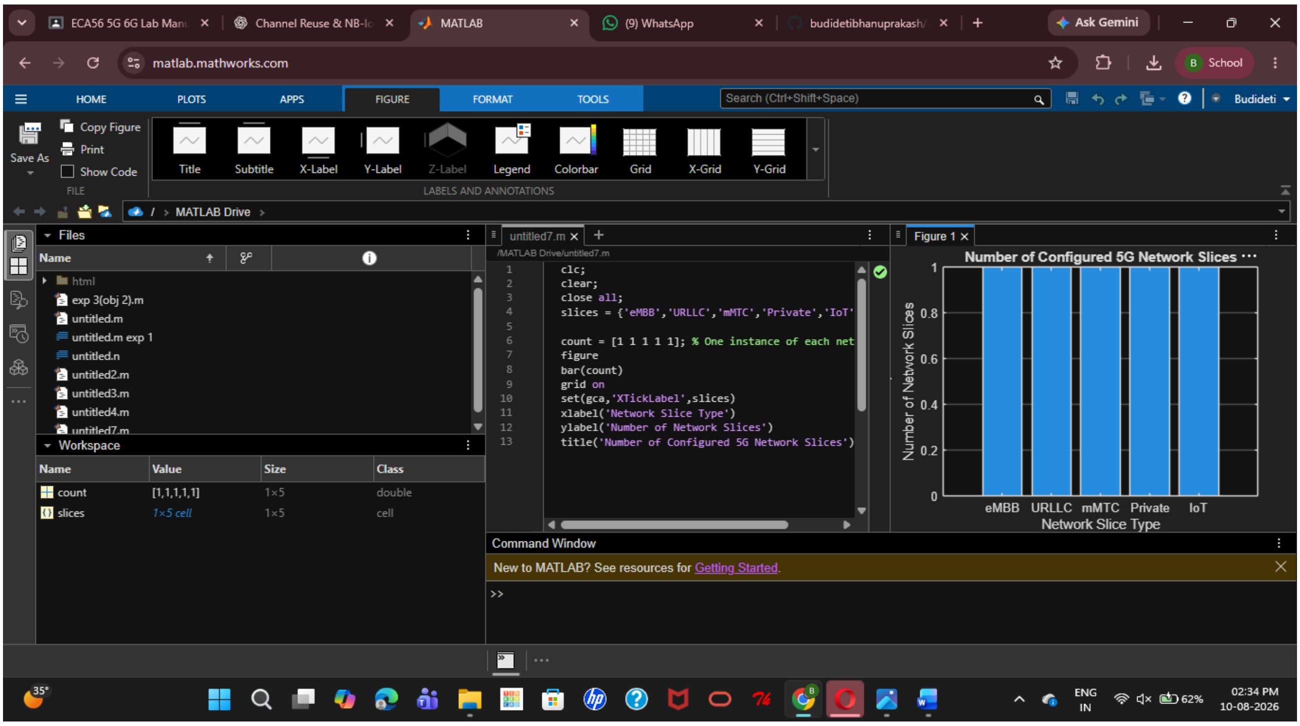
Aim: To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objective 1:

Program:

```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
count = [1 1 1 1 1]; % One instance of each network slice
figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```

Output:



Objective 2:

Program:

clc;

clear;

close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

utilization = [85 70 60 75 50];

figure;

bar(slices, utilization);

grid on;

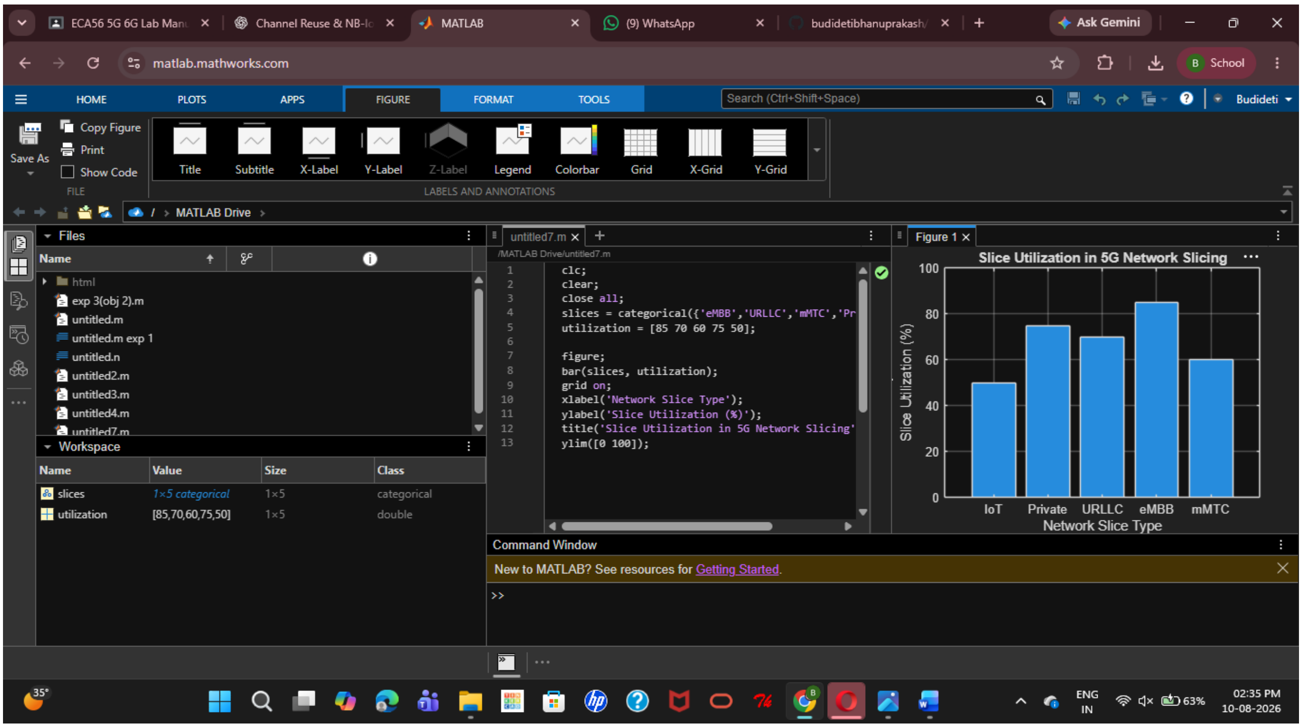
xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);

Output:



Objective 3:

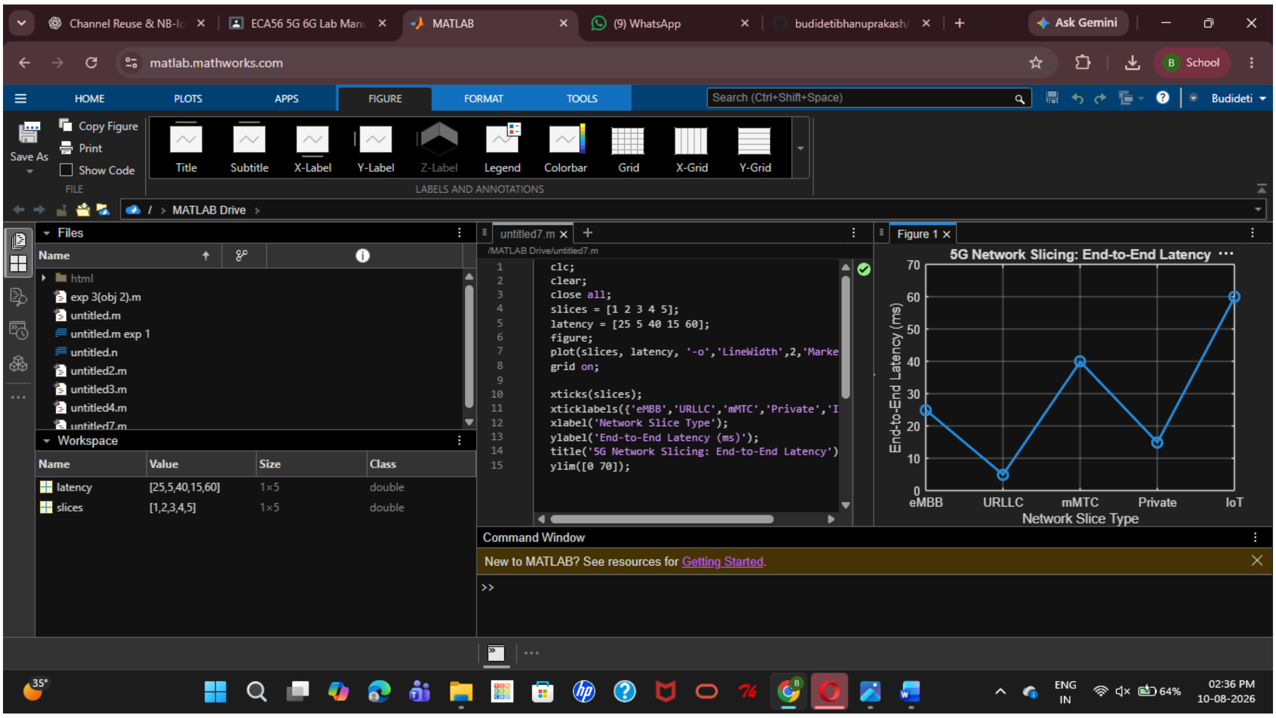
MATLAB Code:

```
clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;


xticks(slices);

xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

xlabel('Network Slice Type');

ylabel('End-to-End Latency (ms)');
```

title('5G Network Slicing: End-to-End Latency');

ylim([0 70]);

Output:



Result:

The 5G network slicing environment was successfully analysed in MATLAB, showing that multiple slices can be efficiently allocated, resource utilization can be compared, and end-to-end latency varies according to the QoS requirements of different 5G services.